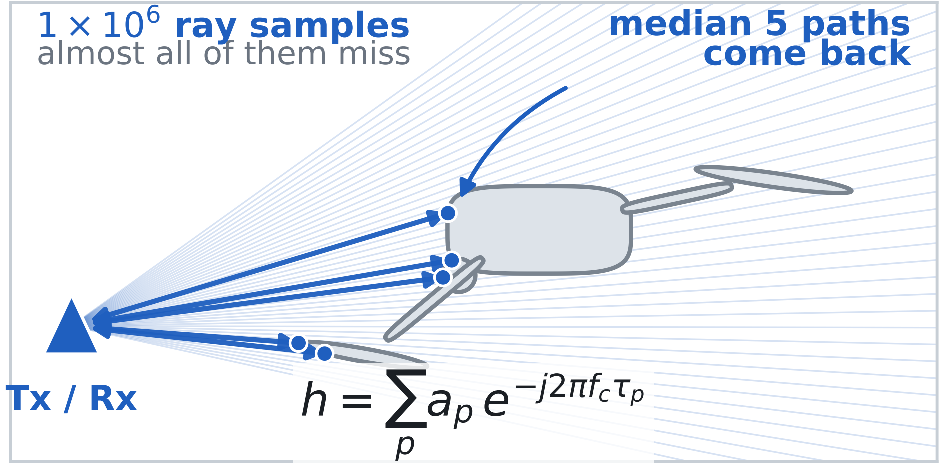
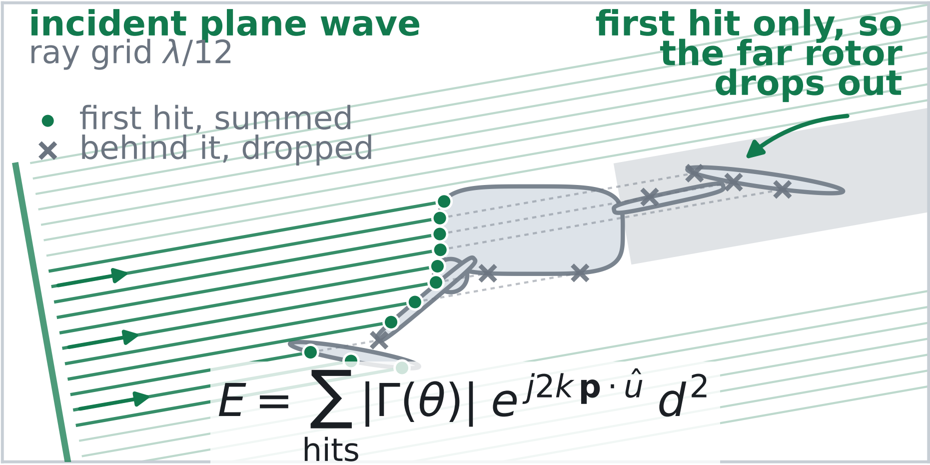
What each of the three engines actually computes

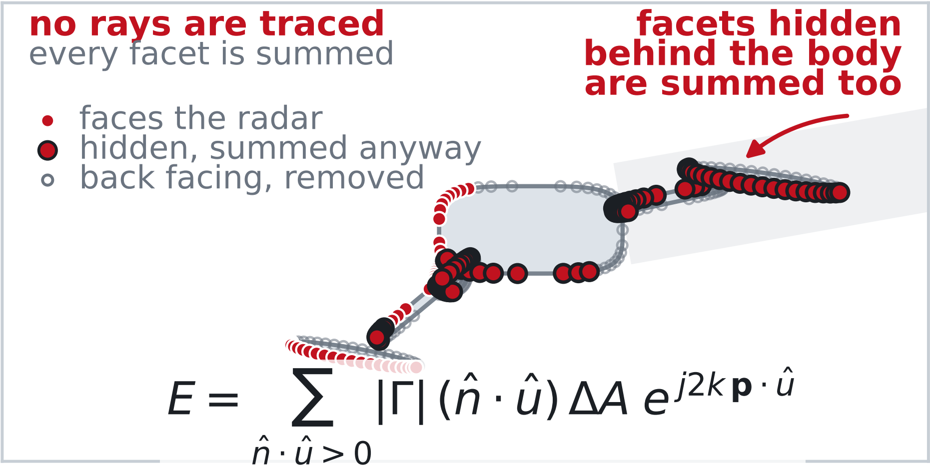
Sionna PathSolver



Ours (SBR+PO, default)



Ours, nothing blocked (control)



	Sionna PathSolver	Ours (SBR+PO, default)	Ours, nothing blocked (control)
FIRED	ray samples fired from the Tx	plane wave on a $\lambda/12$ ray grid, 11k to 16k rays per pose	nothing is fired, no ray tracing
SUMMED	the paths that come back to the Rx	the first hit of every ray	every facet of the same mesh, body and rotors alike
LEFT OUT	the surface integral over the target	back facing and hidden facets	back facing facets only

EVERY COLUMN THEN GOES THROUGH THE SAME SLOW TIME CHAIN

one complex number per rotor pose → 4096 poses on one slow time grid, PRF 19.7 kHz = 208 ms → STFT, 70 sample Hann segments = 0.45 blade periods → the micro-Doppler map, time against Doppler

- Schematic of the three kernels, not a computed result. The airframe is a readable silhouette rather than the simulated mesh.
- The middle column and the control differ in more than occlusion. They also differ in discretisation, a $\lambda/12$ ray grid against surface point clouds at $\lambda/11$ on the blades and $\lambda/6$ on the frame, and in the angle dependent $|\Gamma(\theta)|$, which is on in the middle column and off in the control. Their gap is therefore not the price of occlusion alone. The $(\hat{n}\cdot\hat{u})$ factor appears only in the control because the ray grid already samples the projected area, so both columns are the same physical optics integral.
- The left column is lit by a point source at 3 m while the other two ride an ideal plane wave anchored on the target. The battery and the board sit inside the plastic shell, so the control counts them at full amplitude because it never tests occlusion, the middle column keeps them by tracing through the shell with a two way transmission factor, and the Sionna column never reaches them because refraction is off.